Vienna Motor Symposium 2025

46th International Vienna Motor Symposium
14 - 16 May 2025

2025-32

Is 30 Bar Mean Effective Pressure the Limit for Spark Ignited Commercial Hydrogen Engines?

Ist bei 30 bar Mitteldruck das Limit für fremdgezündete Nutzfahrzeugwasserstoffmotoren erreicht

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DOI: https://doi.org/10.62626/3ce4-qcpb

This paper has been presented and published on the occasion of the 46th International Vienna Motor Symposium 2025.
All papers of this symposium are published in an anthology (ISBN 978-3-9504969-4-9) and can be ordered from the Austrian Society of Automotive Engineers (https://vienna-motorsymposium.com, email: info@oevk.at).
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Abstract

The occurrence of uncontrolled combustion phenomena is the limiting factor for the performance of hydrogen engines. Homogeneity of the air/fuel mixture, avoidance of hot zones in the combustion chamber, appropriate layout of the ignition system and lubrication oil formulation are decisive factors for avoiding combustion anomalies. To optimize all these characteristics, a special engine design would be preferable, with maintaining a high level of commonality with existing diesel engines being the key to market success.

Against this background, FEV has developed a combustion concept with the base diesel engine and its flat cylinder head design that meets the requirements for high charge movement with maximum communality.

The simulations carried out beforehand are compared with the test bench results acquired afterwards for different cylinder head configurations, with a special focus on the pre-ignition tendency. The improved cylinder head design in combination with an improved piston ring design, a special engine oil formulation and an increased injection pressure enables a significant increase in the possible performance level, achieving a mean effective pressure level of 28 bar without and 30 bar with tolerably low pre ignition events.

Kurzfassung

Verbrennungsanomalien sind der begrenzende Faktor für die Leistungsdichte von Wasserstoffmotoren. Gemischhomogenität, Vermeidung heißer Zonen im Brennraum, geeignete Auslegung des Zündsystems und Schmierölformulierung sind die entscheidenden Faktoren, um diese Grenze zu verschieben. Um all diese Eigenschaften zu optimieren, wäre eine spezielle Motorkonstruktion wünschenswert, wobei eine hohe Gleichteileanzahl mit den bestehenden Dieselmotoren der Schlüssel für den Markterfolg sein wird.

Vor diesem Hintergrund hat FEV ein Brennverfahren entwickelt, das die Anforderungen an eine hohe Ladungsbewegung bei maximaler Übereinstimmung mit dem Diesel-Basismotor und seinem flachen Zylinderkopfdesign erfüllt. Die im Vorfeld durchgeführten Simulationen werden mit den später durchgeführten Prüfstandsergebnissen für verschiedene Zylinderkopfkonfigurationen unter besonderer Berücksichtigung der Vorentflammungsneigung verglichen. Das verbesserte Zylinderkopfdesign in Kombination mit einem optimierten Kolbenringdesign, einer speziellen Motorölzusammensetzung und erhöhtem Einspritzdruck ermöglicht eine signifikante Steigerung des Leistungsniveaus auf ein Mitteldruckniveau von 28 bar ohne und 30 bar mit geringem Auftreten von Vorzündungen.

Introduction & Motivation

Given the high technological maturity level of internal combustion engines (ICEs) and the gravimetric energy density of hydrogen compared to batteries, there is a continuously increasing interest in internal combustion engines powered by hydrogen as a decarbonization solution for the on- and off-road market. These engines typically run as mono-fuel engines on the principle of premixed combustion with spark ignition, which is different from the diffusive combustion of the base diesel setup. Therefore, important decisions must be made in the architecture strategy of combustion engines for operation with hydrogen (see Figure 1).

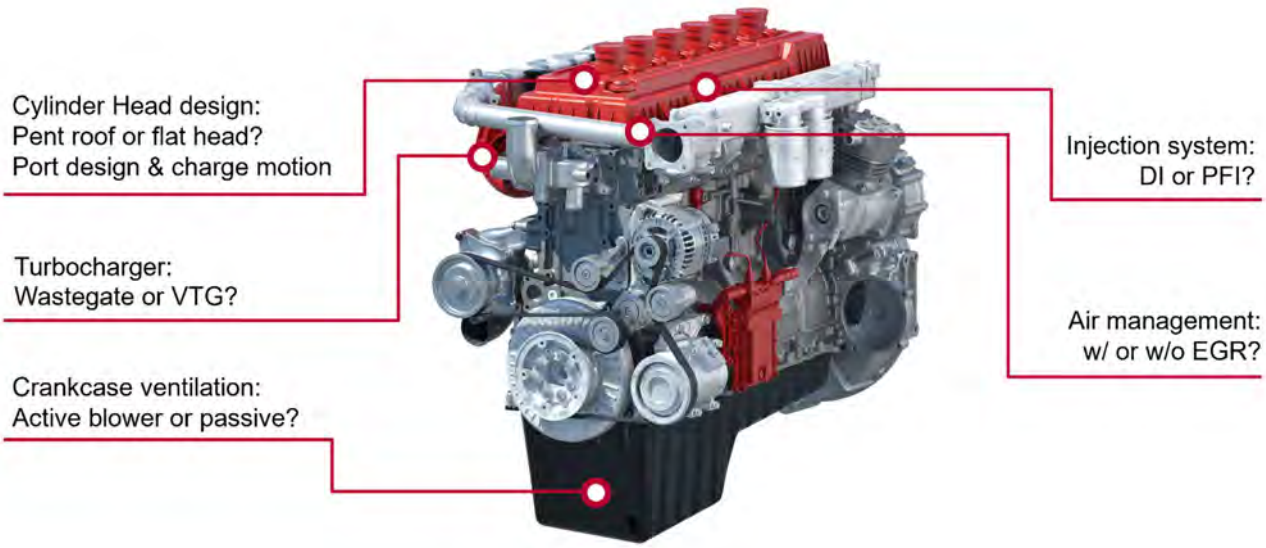

Figure 1: Decisions regarding architecture strategy for alternative fuels

The biggest challenge in achieving performance parity with diesel engines is the occurrence of uncontrolled combustion phenomena (pre-ignition and knocking) with fuels with low required ignition energy like hydrogen. In this context, the homogeneity of the air/fuel mixture, the turbulence to increase mixing and flame speeds and the avoidance of hot zones in the combustion chamber are the key elements to shift the limit to higher power density. The decision about the cylinder head & port design is crucial in achieving these objectives. Moreover, the appropriate layout of the boosting system to achieve target charge dilution is necessary. The ignition system and the lubrication oil formulation are also decisive factors for avoiding combustion anomalies.

One of the other decisive decisions must be taken on the injection system. Port fuel injection (PFI) systems are being utilized in series production engines for hydrogen since 2024 but if highest power density is targeted, the drawback on volumetric efficiency needs to be avoided. In 2025, the first H_2 direct injection (DI) engines are announced to be installed in a small fleet of trucks [1]. The direct injection of hydrogen eliminates the loss in volumetric efficiency compared to PFI systems but creates a tough challenge of sufficient mixture homogenization which needs to be tackled by tailored charge motion design. The most effective way to improve mixture homogenization would be a pent roof cylinder head design

to significantly increase level of charge motion but requires a complete redesign of the cylinder head.

Since the introduction of hydrogen powered internal combustion engines will ramp up gradually during the next years, high communality with today`s diesel engines and the capability to produce these engines on the same manufacturing line are key. Therefore, the design of cylinder heads of hydrogen combustion engines and the mixture formation concept is dominated by the degree of targeted communality with the base engine. Typically, this layout and optimization is done using 3D CFD simulations, which model the in-cylinder flow considering the intake and exhaust ports as well. Such simulations can also model the hydrogen injection in combination with the combustion process. However, these simulations can be cost and time intensive especially when they consider the direct injection and multiple cylinder head concepts. Additionally, they require detailed inputs regarding the operation and injection parameters.

FEV's Charge Motion Design (CMD) process provides an effective solution to optimize the head, port and combustion chamber design for hydrogen combustion while in parallel considering potential benefits also for a multi-fuel concept during the initial development stage. The CMD process accelerates the development and provides a cost-effective solution by combining CFD simulations, benchmark data and fuel specific correlations to rank the design variants and give first impressions of the combustion performance in the layout phase. The evaluations can then be updated for the specific fuel and injection concept during the detailing phase when the required inputs for 3D CFD simulations are available. This paper describes the CMD process and highlights its application to evaluate different head concepts for a heavy-duty engine to extend the current limitation to and beyond 30 bar mean effective pressure level.

Base engine design

The charge motion design of a conventional commercial diesel engine aims to produce an efficient and turbulent airflow that maximizes fuel-air mixing for better combustion. This involves controlling swirl and the motion of the air through the combustion chamber. While this principle evolved during the last decades, the requirement of spark-ignited combustions systems, and in combination, the usage of directly injected gaseous fuels hasn´t been in focus in the past for engines for commercial vehicle application. The level and orientation of the charge motion in a hydrogen combustion engine are influenced by three variables: intake port design, piston shape and injector positioning (spray cap orientation). To fulfil the requirement of a high degree of communality with the base diesel engine, the scope for optimizing these three parameters is limited. Therefore, the usage of the well-established pent roof cylinder head design known from gasoline engines for decades is not possible in many cases. In order to achieve a sufficient level of tumble charge motion, shrouds (small plates with asymmetric cross section which sit above the intake seat ring) can be applied to mask a certain area of the intake port and therefore guide the incoming charge air flow towards the cylinder head and consequently create tumble based charge motion (see Figure 2). The usage of shrouds can be beneficial if the design of a dedicated tumble orientated intake port is not possible. However, these shrouds result in a volumetric efficiency penalty for a targeted tumble level compared to a dedicated tumble-focused port design, thus reducing the air fuel ratio at full load or reducing the maximum load level.

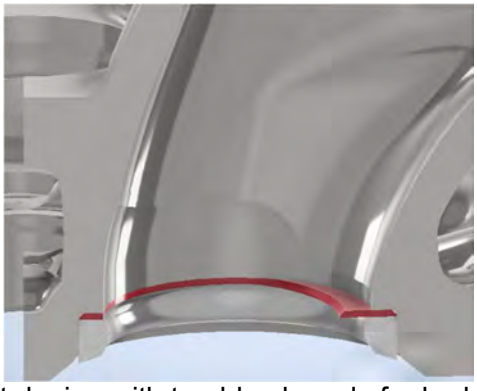

Figure 2: FEVs filling port design with tumble shrouds for hydrogen commercial engine

To overcome these drawbacks, FEV has designed and optimized a dedicated tumble port and implemented it into a state-of-the-art diesel engine cylinder head design maintaining the degree of commonality in terms of machining processes (valve seat and valve guide cutting) as shown in Figure 3.

Figure 3: FEVs tumble port design for hydrogen commercial engines

The manufacturing steps for injector and spark plug sleeve installation will need to be individualized for diesel and hydrogen combustion engines. To tackle the challenging task of port design under the given restrictive boundary conditions, FEV used the self-developed Charge Motion Design (CMD) process, which is described in the following chapter.

Charge Motion Design (CMD) Process

The target of the Charge Motion Design (CMD) process is the optimization of flow-guiding surfaces of the combustion system, injector layout, and combustion properties (see Figure 4) for premixed combustion concepts. This process strongly focuses on in-cylinder 3D-CFD simulations to narrow down and rank the hardware variants to minimize the testing efforts. These simulations are then post-processed with proprietary scripts that generate an array of direct and analyzed variables considering the in-cylinder flow and turbulence. These are then plotted into multiple correlation scatter-bands, which consider not only the port and combustion chamber geometry dimensions, but also the charge motion and turbulence parameters. These scatter-bands form an essential component of the CMD process and allow the evaluation of a given design quickly and in relation to the benchmark based on decades of engine development at FEV. Furthermore, the underlying database is used to directly identify required geometry changes.

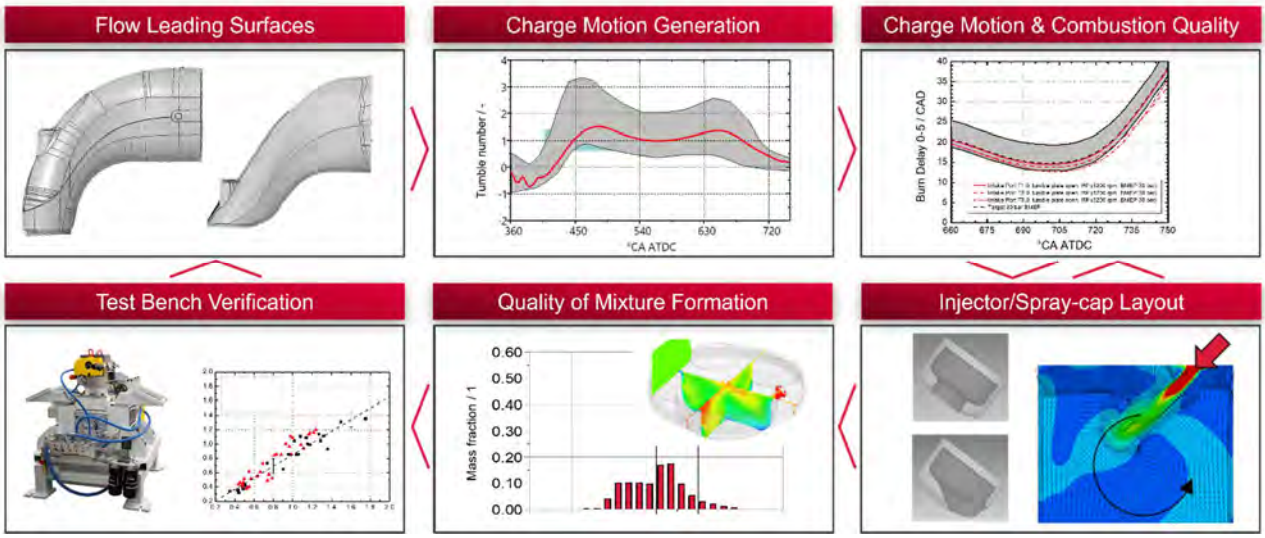

Figure 4: Charge motion design (CMD) process at FEV

For premixed combustion systems, a high tumble level is the preferred approach to enable efficient and stable engine operation. The tumble from the inflow is converted into turbulent kinetic energy close to the top dead center (TDC), which further supports the mixing process and accelerates flame propagation. Thus, the target of the CMD process is not only to generate a high initial charge motion from the combustion chamber and port layout, but also to ensure good conservation of tumble motion and conversion into turbulent kinetic energy (TKE).

Another optimization target within the CMD process during the detailing phase is the injector layout. High-precision injector models have been developed at FEV and validated against optical measurements for a variety of fuels, including gasoline, CNG, hydrogen, and methanol [2, 3]. The target for the injector layout is the reduction of wall impingement and the overall improvement of mixture homogeneity. For lean operation, the stratification within the cylinder is of particular importance because NO_X emissions must be minimized. The interaction of the injection with the charge motion is modelled and optimized with CFD simulations.

Since TKE and characteristic numbers for fuel homogeneity do not correlate directly with the combustion properties, FEV has developed two models to predict the combustion performance of the engine. The first model evaluates the burn delay (0-5 % mass fraction burned) during the early stage of combustion [4]. The second model evaluates the 5-50 %

mass fraction burned by correlating burn duration with a turbulent characteristic number [5]. Fuel-specific parameters and flame speeds are implemented in both models. The knowledge of the burn delay provides an evaluation of the conditions at the spark plug. It can detect when the optimum is reached and whether it is sufficient for stable ignition. A faster 5-50 % burn duration can be directly correlated to the efficiency of the engine or leveraged to work with higher air/EGR rates. Thus, the optimization of these two stages of combustion is necessary.

RANS CFD simulations were performed at a high load operating point (1600 rpm, 24 bar IMEP) with hydrogen direct injection for the SCE setup. The injector was modelled considering the inner valve geometry and imposing the mass flow rate and needle motion. The three hardware variants mentioned in the previous section (base ports, base ports with tumble shrouds and dedicated tumble ports) were simulated and the optimum defined for the shrouds and the tumble ports. A pot bowl was used for these simulations as this shape is realizable in hardware in the base piston design while allowing sufficient space for the piston cooling. The results of the gas exchange simulation are plotted in the charge motion scatter-bands in Figure 5, where the shaded region shows the benchmark engines with bore > 90 mm.

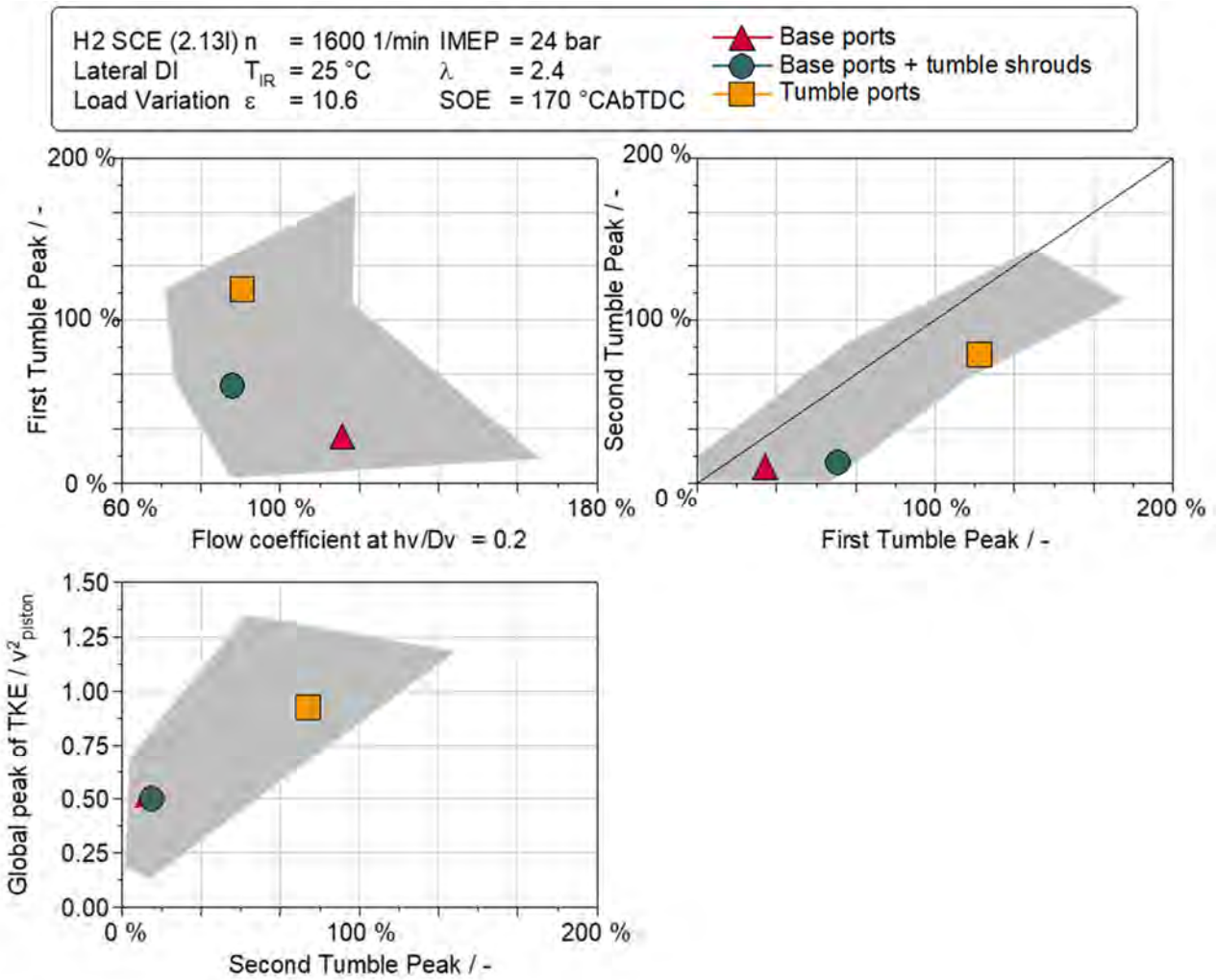

Figure 5: CMD Process scatter-bands based on results of the CFD simulations

In the first trade-off between flow coefficient and tumble, it can be observed that the flow performance must be sacrificed to create tumble motion. Stationary CFD simulations were performed to calculate the flow coefficients for the three concepts. The tumble ports generate significantly higher tumble compared to the shrouds for a given drop in flow coefficients. The base port design produces no charge motion during the gas exchange process. The hydrogen injection only produces a weak cross tumble motion, which is not supported by the pot bowl piston design. By inserting the tumble shrouds into the intake port, the tumble peak is significantly increased during the intake valve event. However, this tumble level quickly disappears during the compression stroke. The dedicated tumble ports generate a tumble peak during the intake stroke twice higher than the shrouds. Consequently, despite the pot bowl, some of the tumble motion is conserved into the compression stroke. However, these flat roof concepts are still lower than other pent-roof concepts in the benchmark, which produce higher tumble peaks and can better maintain the tumble thanks to lens-shaped bowl designs. The turbulence generated during the compression stroke through the destruction of the tumble, is directly proportional to the second tumble peak as shown by the shaded plot. Only the flat head tumble port design has a significant level of turbulence coming from the moderate second tumble peak.

The mixture distribution at TDC coming from the 3D CFD simulation with hydrogen injection is shown in Figure 6. The normalized standard deviation of the equivalence ratio is chosen to represent the homogeneity in the combustion chamber. At this load point (1600 rpm, 24 bar IMEP) with chosen injector configuration, only a small improvement can be observed in the homogeneity by the introduction of the tumble shrouds. The optimized tumble ports improve the homogeneity including the small volume in front of the injector. Consequently, the tumble port design is chosen and will be manufactured for the next measurement campaign.

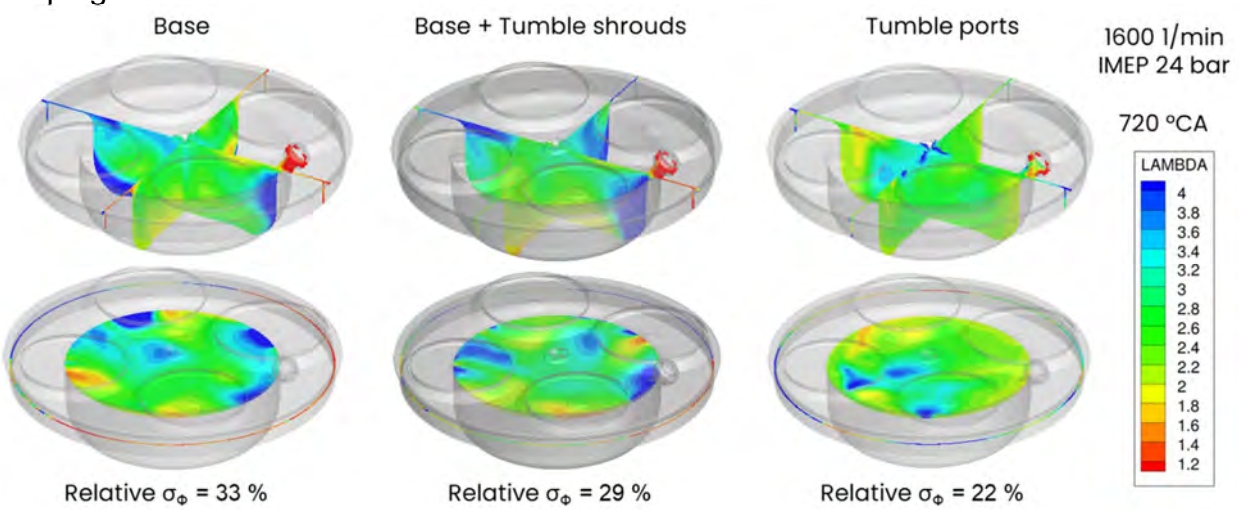

Figure 6: Results of mixture formation CFD simulation

The burn duration correlations for 0 to 5 % and 5 to 50 % were run for the three concepts and are plotted in Figure 7. The spark timing is varied in the post processing and plotted on the X-axis and the total burn duration from spark timing to 50 % is plotted on the Y-axis.

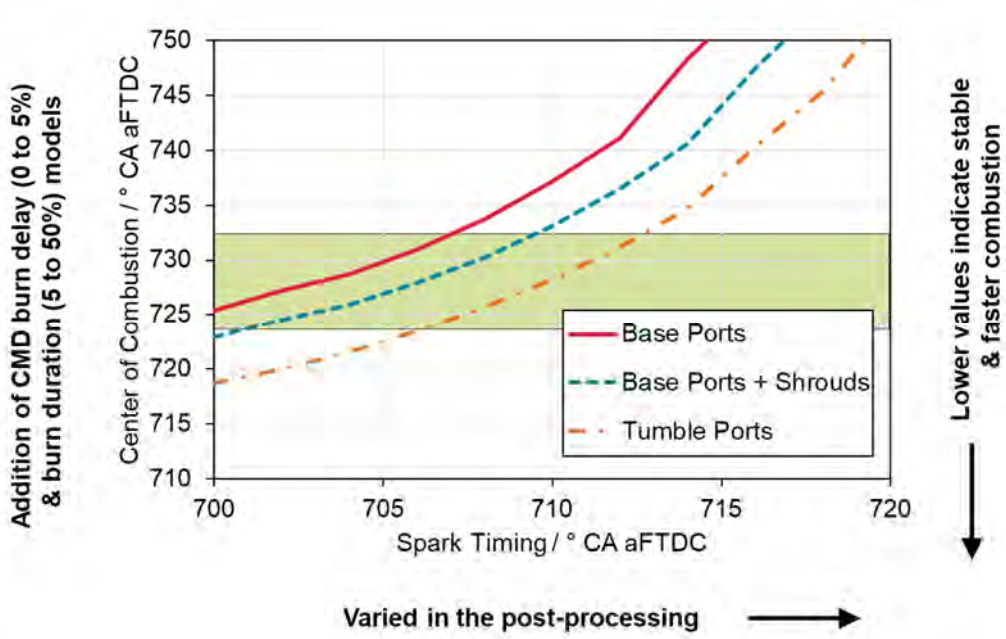

Figure 7: CMD burn duration evaluation for chosen concepts.

A lower lying curve indicates a faster and stable combustion. In this regard, the tumble shrouds already show an advantage over the base port design and the tumble ports are even better.

Test bench setup and results

For the experimental investigations, a heavy-duty diesel single cylinder engine was modified to a pre-mixed hydrogen combustion system. A pot bowl piston design has been chosen to achieve a compression ratio of 10.6:1 with the flat cylinder head design of the base diesel engine. The cylinder head was redesigned to implement the low-pressure direct injection of hydrogen in a lateral position, enabling an injector swap without cylinder head disassembly to ensure high flexibility in the selection of components. To fulfill the requirement of high communality with the base diesel engine, the valve position is carried over. The base engine intake ports provide maximum cylinder filling with a very low level of charge motion. To increase the level of charge motion, tumble shrouds were implemented upstream the intake valve seats.

During all measurements, the exhaust backpressure was controlled to be equal to the boost pressure. Unless otherwise stated, a high-flow low-pressure injector at a rail pressure of 30 bar was used in all measurements. The intake air was dried to a relative humidity of 3 - 10 % depending on air mass flow. A summary of the engine specifications and further boundary conditions for operation can be found in Table 1.

Description	Unit	Value
Bore x stroke	mm	132 x 156
Displacement	cm^3	2135
Compression ratio	-	10.6:1
Coolant / oil temperature	°C	90 / 90
Temperature in intake runner	°C	25

Injection pressure (DI)	bar	30
Spark Plug	-	Cold heat range
Charge motion	-	None / Tumble
Injector	-	High Flow LPDI

Table 1: Engine characteristics and boundary conditions for operation

To reach high specific power, abnormal combustion must be avoided. For a DI engine, knock and pre-ignition (PI) are the relevant phenomena, as backfire cannot occur with a start of injection after intake valves close. The occurrence of abnormal combustion events is depending on the ratio between required ignition energy of the mixture and the provided energy during intake and compression stroke.

The energy supply can be controlled on the one hand by optimized hardware to minimize oil transport to the combustion chamber and to avoid hotspots, but also by adapted boundary conditions such as lower coolant, oil and intake air temperatures. The required ignition energy of the mixture on the other hand is heavily dependent on gas composition, which can globally be altered by air fuel ratio (AFR), exhaust gas recirculation (EGR) and water injection. On a smaller scale, charge motion is relevant to avoid local rich zones and residual gas pockets, as these zones are prone to cause abnormal combustion [6, 7, 8].

In the following diagrams, the pre-ignition frequency is displayed in the top left diagram, average knock amplitude and the maximum knock amplitude are shown in the top right graph. In the bottom left, the indicated specific NO_x ($ISNO_x$) emissions can be seen and lastly the efficiency is plotted on the bottom right. The X-axis is either indicated mean effective pressure (IMEP) or the center of combustion/point of 50 % heat release (α_{50}) for the spark advance variation.

In the following diagrams, the relevance of charge motion can be observed. At the lower engine speed of 1200 1/min (see Figure 8), the tumble configuration (base port + shroud) shows lower PI rates than the base configuration. At 1600 1/min (see Figure 9) the achievable load without tumble increases as the charge motion (generated by the hydrogen injection) gets more intense, and in parallel the cycle duration shortens. Consequently, the time a given volume element is exposed to a hot surface area is reduced by a shortened cycle duration and an increased transport speed within the combustion chamber.

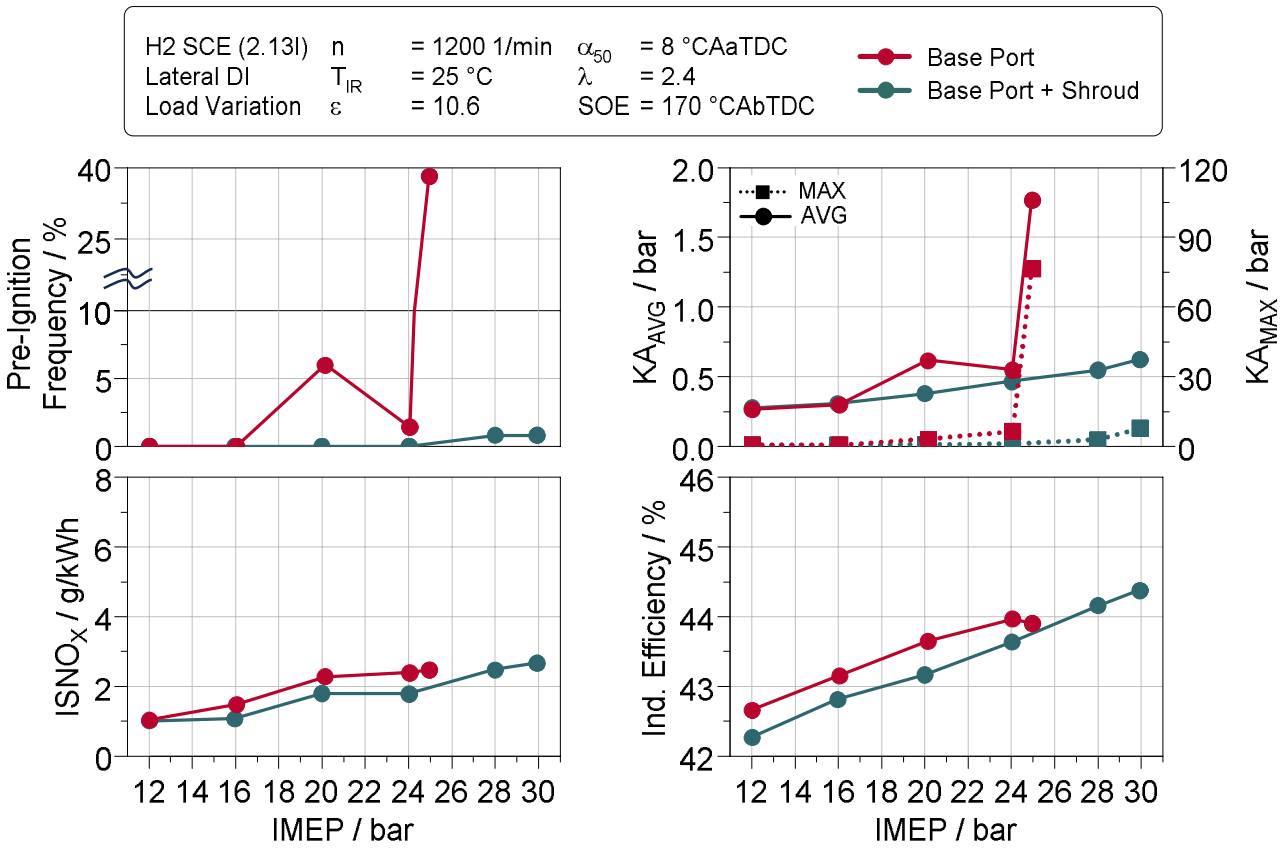

Figure 8: Load sweep Base Port vs Tumble Shroud, high flow injector, 1200 1/min

The base configuration also shows significantly more frequent knock with high amplitudes beyond 24 bar IMEP at 1200 1/min. The sudden increase in knock shows the challenging operation close to the knock limit for hydrogen engines and the necessity to improve the robustness against abnormal combustion. Knock is influenced by mixture homogeneity and peak temperatures during combustion. At high PI rates, temperatures rise as the combustion chamber heats up due to high peak pressures and temperatures, potentially causing more knock, which in turn causes more pre-ignition, resulting in a self-propelled cycle of abnormal combustion. Charge motion has the same effects for knock as for PI, because rich zones are reduced, and increased velocities allow less time for energy transfer.

The improvement in mixture homogeneity causes lower NO_x emissions for the tumble setup with similar efficiency level. At high PI and knock frequencies, efficiency decreases because combustion occurs before TDC in some cycles, resulting in higher compression and wall heat losses.

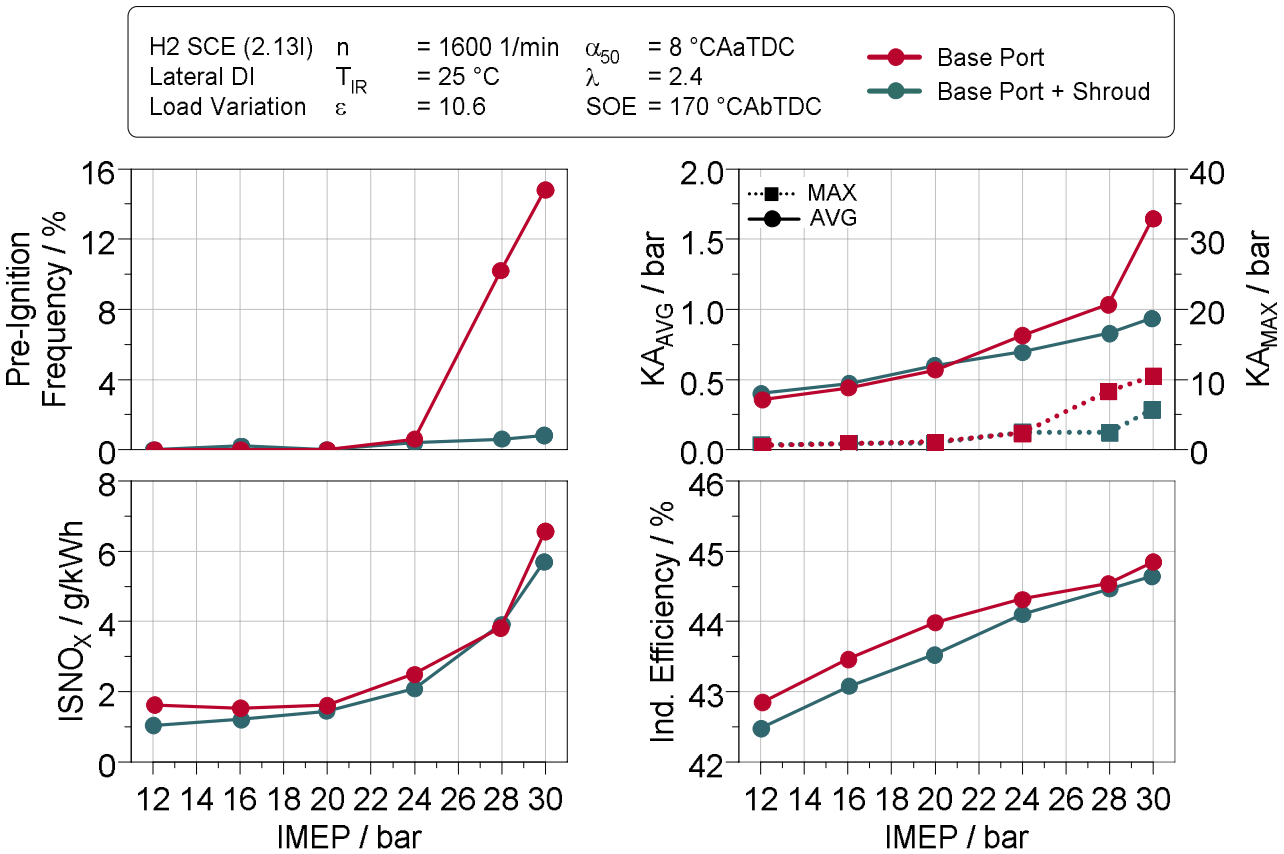

Figure 9: Load sweep Base Port vs Shroud, high flow injector, 1600 1/min

Figure 10 shows the importance of an injector with a sufficient flow rate and the capability to handle higher rail pressures. On the one hand, the achievable maximum load is reduced with low pressure and flow rate as the injection duration is limited by the pressure ratio between combustion chamber and injector nozzle. On the other hand, the mixture formation suffers with a lower injection pressure, as the spray penetration decreases, the time for mixture formation shortens and the interaction with charge motion is reduced. This leads not only to slightly higher PI frequencies but also to significantly increased knock amplitudes already at IMEP = 24 bar for the regular injector setup. As the duration of sub-critical flow during injection increases compared to the high flow injector, leading to an increase in rich zones, NO_X emissions also increase more. Furthermore, the freedom in operation strategies increases with high flow injectors, as they cannot only operate at higher pressure to enable stratified combustion for increased efficiency in part load operation, but also have short injection durations at the start of compression to provide maximum time for mixture formation.

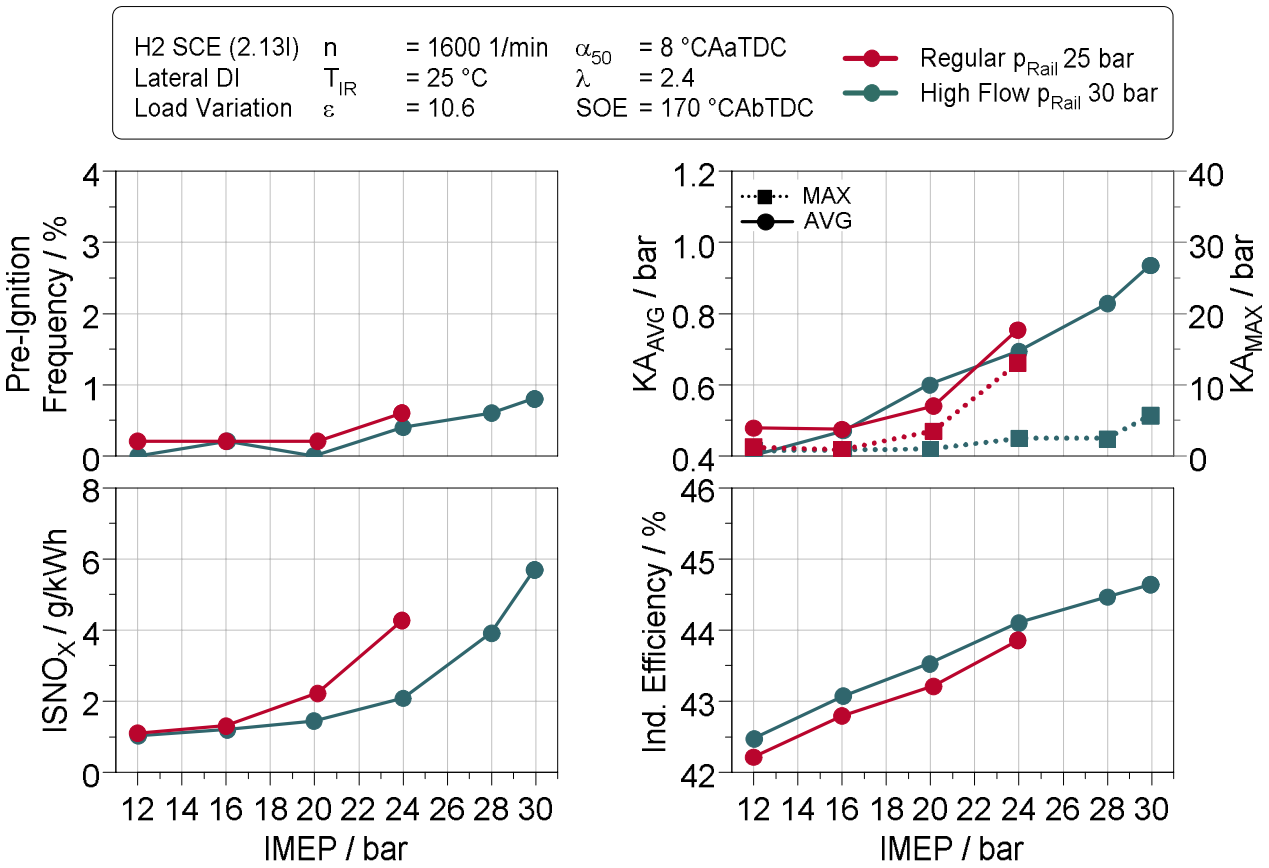

Figure 10: Load sweep regular vs high flow injector, Tumble shrouds, n = 1600 1/min

For hydrogen combustion, the window of ideal spark timing is significantly limited by abnormal combustion. To mitigate knock in a gasoline engine, the retardation of spark timing is a sufficient countermeasure. For hydrogen, however, the retardation is limited by the occurrence of higher PI rates. Advancing spark timing to improve efficiency by taking advantage of the fast combustion speed of hydrogen is limited by a sudden increase in knock frequency and amplitude. In Figure 11 a variation of spark timing with higher and lower intake temperatures are compared showing that the occurrence of knocking combustion is more direct dependent on temperature than pre-ignition. With retardation of the spark timing. the PI frequency increases. This is caused by two effects. Firstly, there is more time to initiate pre-ignition. For this engine, the start of PI is usually around 8-10 °CA bTDC. As spark timing is usually more advanced, the combustion is faster than most of the potential PI events. With retardation, the ratio of cycles that show pre-ignition increases. The second effect is an increasing residual gas temperature with potentially higher hydrogen concentration. Furthermore, the exhaust gas temperature rises and leads to higher surface temperatures promoting energy transfer from wall to gas. The exact influence of residual gas composition and temperature are planned to be investigated further.

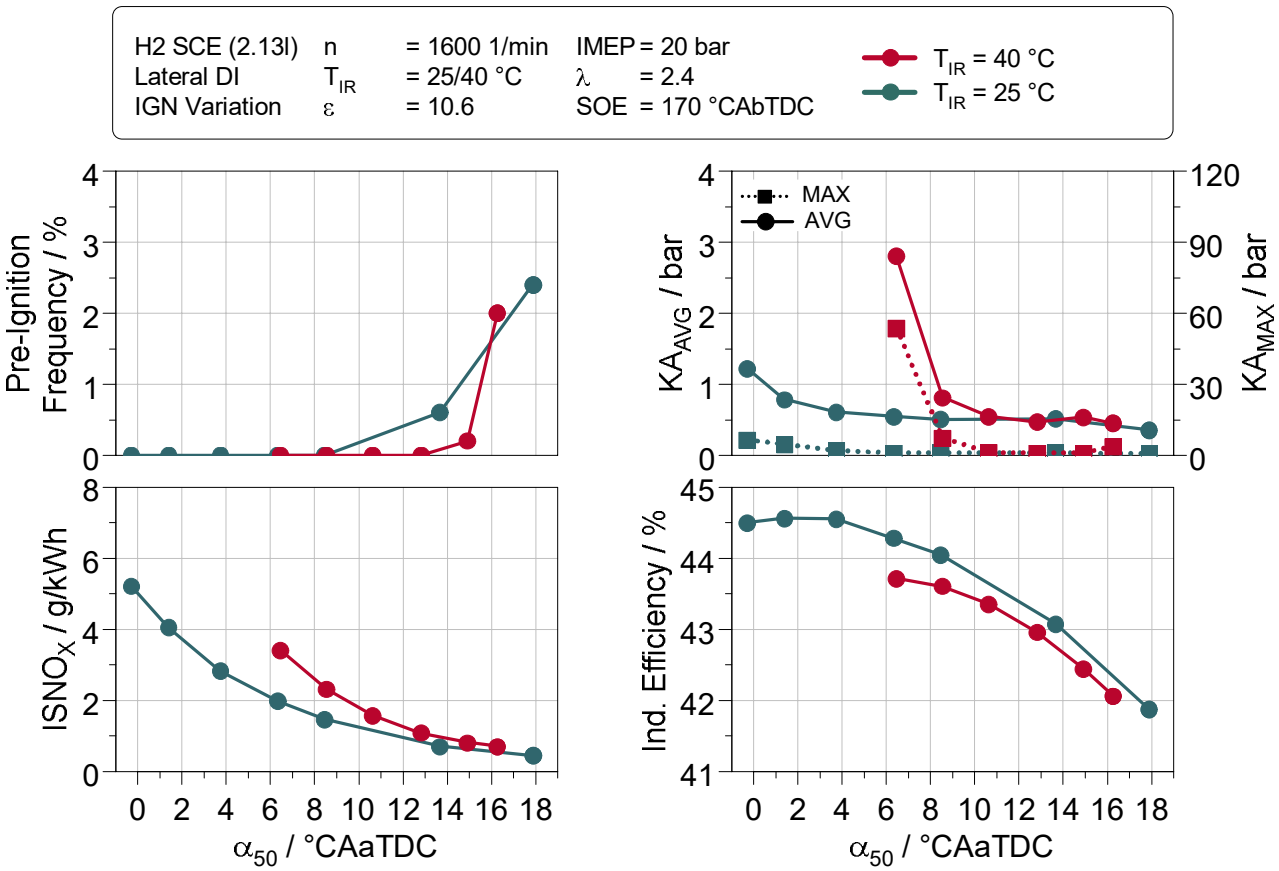

Figure 11: Ignition variation, high vs low intake temperature, 1600 1/min

This variation of intake air temperature shows the potential of improved charge air cooling to reduce anomalous combustion. To increase the required ignition energy, it is also possible to raise the water content in the intake air. Investigations with a humidity simulation will be conducted in the future. Additionally, it is possible to add water injection into the intake runner to take advantage of the water's evaporation enthalpy to reduce the temperatures inside of the cylinder.

The addition of high-pressure external exhaust gas recirculation is also possible but is less desirable as water can condense in the EGR cooler and intake runner, so measures must be taken to direct the water flow into the cylinder and to separate it when the engine is shut down. In previous studies, NO_X in the exhaust gas was found to promote knock in gasoline engines, an effect that requires further investigations for hydrogen engines. [9,10]

To determine the effect of combustion anomalies on engine health, an endurance run with over 1,000,000 engine cycles has been conducted. The engine was operated in high load conditions at rated speed to maximize the strain on the engine. Three different spark timings were investigated, representing a center of combustion of 8 °CA, 12 °CA and 20 °CA aTDC. The focus was set on the latter two timings to maximize pre-ignition occurrence. The histograms of the peak pressure and knock amplitude are displayed in Figure 12.

As expected, the highest average peak pressure happens at the earliest ignition timing, but maximum peak pressures over 230 bar mostly occur at later ignition timings due to pre-ignition. Therefore, the ignition angle needs to be controlled in a way to minimize excessive stress on the engine. A large number of pre-ignition events resulted in high knock amplitudes even at late ignition angles.

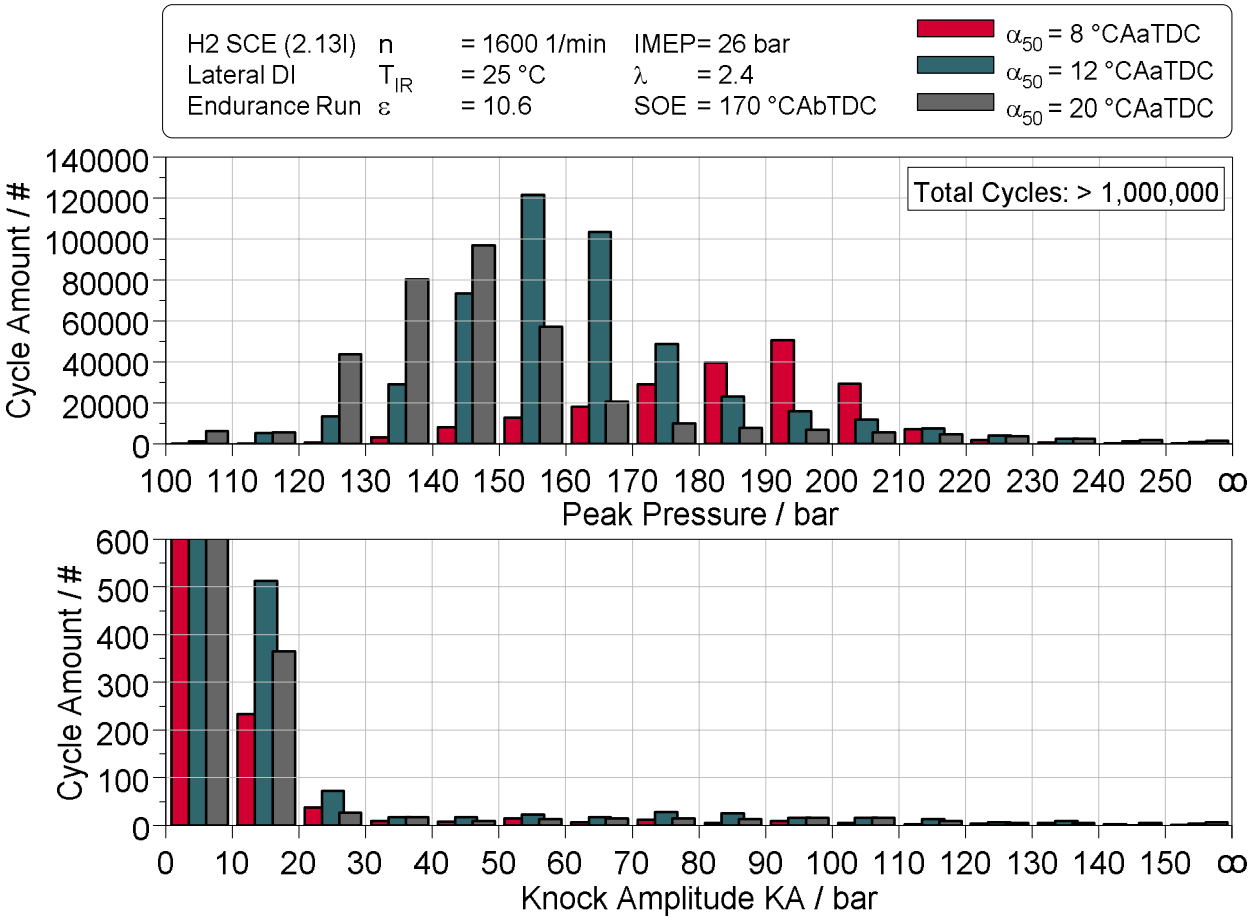

Figure 12: Histogram of peak pressures and knock amplitudes in three operating points over 1,000,000 cycles

In total, more than 1,700 engine cycles exhibited knock amplitudes above 10 bar while around double the amount displayed amplitudes between 5 and 10 bar (not shown in diagram). This excessive strain on the engine did not result in any damage to the cylinder head or piston. Extrapolating the number of pre-ignition events with a PI rate of 0.5 % to the engine lifetime (1,200,000 km @ 80km/h, 15000h, 1% full load) would result in a total number of pre-ignitions of 27,000. This number of pre-ignitions is 16 times higher than tested up to now. However, the next durability run is planned to demonstrate a permissible pre-ignition rate of 0.5 % over engine lifetime.

Conclusions and Outlook

For a pre-mixed hydrogen combustion system, which is dominated by its tendency towards pre-ignition and knocking combustion, achieving diesel-like power levels on a heavy-duty engine is a great challenge. To achieve maximum performance, mixture formation and turbulence in the combustion chamber must be optimized.

FEVs Charge Motion Design (CMD) process combines the usage of CFD simulations and benchmark scatterbands to evaluate and optimize the cylinder head concept designs. Burn delay and burn duration correlations for different fuels based on a single turbulent characteristic number enable predicting the performance of the layout. Special attention is paid to the propagation of hydrogen flames due to their strong wrinkling. Tumble shrouds

and tumble ports, which showed better homogeneity and faster and stable combustion were defined and optimized using the CMD process.

By introducing tumble motion into the combustion chamber, the dwell time of hydrogen near local ignition sources is reduced. This, in combination with an improved mixture homogeneity, leads to a significant decrease in pre-ignition occurrence, which enables stable operation at maximum loads of 30 bar IMEP. Further improvement of the mixture homogeneity was achieved by using a high flow injector. The increased flow rate gives the mixture more time to homogenize, resulting in an overall improvement in engine performance and emissions. Aside from the load, the spark timing also heavily influences abnormal combustion phenomena. At an intake air temperature of 40 °C, centers of combustion before 8 °CA aTDC are not feasible due to knock. Late ignition timings, however, increase the pre-ignition frequency, so the stable operating range is narrow. This was emphasized during an endurance run at high load with over 1,000,000 recorded engine cycles. The most critical conditions were recorded at late ignition timings with a high number of cycles with high peak cylinder pressures and knock amplitudes.

With dedicated and optimized hardware, loads even higher than 30 bar IMEP are achievable. An optimized tumble motion combined with a high flow injector can improve mixture formation, which is key for reducing abnormal combustion. FEV has designed and optimized such a cylinder head concept with the CMD process, which is planned to be tested on a single cylinder engine in the near future. In addition, an efficient charge air cooling system is required to minimize knock occurrence and maximize filling. Finally, the oil formulation and the oil transport into the combustion chamber need to be optimized for hydrogen operation.

Acknowledgements

The authors would like to thank Dr. Max Mally and Mr. Josef Beuel from FEV Europe GmbH for their contributions to this work as well as Hoerbiger AG for supporting with injection equipment.

References

1. P. Albrecht, “Hydrogen Powertrain Solutions for HD-Trucking”, Aachener Kolloqium, 2024.
2. I. Singh, A. Güdden, A. Raut, A. Dhongde, A. Emran, V. Sharma & S. Wagh, “Experimental and Numerical Investigation of a Single-Cylinder Methanol Port-Fuel Injected Spark Ignition Engine for Heavy-Duty Applications”, SAE Technical Paper 2024-26-0072, 2024.
3. A. Dhongde, P. Recker, H. Sankhla, L. Virnich, “Advanced Simulation Methodologies for Hydrogen Combustion Engines”, ATZheavy duty worldwide, 14(4), 46-51, 2021.
4. M. Budde, “Untersuchung zyklischer Schwankungen der Entflammung an der Klopfgrenze von magerbetriebenen Ottomotoren”, Dissertation RWTH Aachen, 2012.
5. W. Wiese, S. Pischinger, P. Adomeit, J. Ewald, "Prediction of Combustion Delay and Duration of Homogeneous Charge Gasoline Engines based on in-Cylinder Flow Simulation", SAE Technical Paper, 2009-01-1796, 2009.

6. Meske, R., Zimmer, P., Virnich, L., et al.: Component and Combustion Optimization of a Hydrogen Internal Combustion Engine to Reach High Specific Power for Heavy-Duty Applications, JSAE Kyoto, 2023
7. Boberic, A. et al.: Measures to achieve high specific power with a heavy-duty H2 internal combustion engine: A numerical and experimental analysis, 31. Aachen Colloquium Sustainable Mobility, 2022
8. Güdden, A., Zimmer, P. et. al.: Performance improvement of direct injection H2-ICE with flat cylinder heads, 13. Dessauer Gasmotoren-Konferenz, 2024
9. Mally, M.: Experimentelle und numerische Untersuchung von gekühlter Abgasrückführung an der Volllast eines aufgeladenen Ottomotors, Dissertation, RWTH Aachen, 2021
10. Kawabata, Y., Sakonji, T., and Amano, T., "The Effect of NOx on Knock in Spark-ignition Engines," SAE Technical Paper 1999-01-0572, 1999, https://doi.org/10.4271/1999-01-0572.